O1 — Organic Chemistry I

1. Nomenclature and Functional Groups

1.1 IUPAC Naming Rules

Organic molecules are named systematically using the IUPAC (International Union of Pure and Applied Chemistry) convention. The name consists of three parts:

1. **Root/prefix:** Identifies the longest continuous carbon chain: meth- (1C), eth- (2C), prop- (3C), but- (4C), pent- (5C), hex- (6C), hept- (7C), oct- (8C), non- (9C), dec- (10C).
2. **Suffix:** Identifies the highest-priority functional group: -ane (alkane), -ene (alkene), -ol (alcohol), -al (aldehyde), -one (ketone), -oic acid (carboxylic acid), -amine (amine), -amide (amide).
3. **Substituents/side chains:** Named with a position number (locant) and prefix. Use the lowest set of locants. Multiple identical substituents use di-, tri-, tetra- prefixes.

Naming procedure:

1. Find the longest continuous carbon chain that includes the principal functional group.
2. Number from the end nearest to the principal functional group (or nearest branch point if no functional group suffix).
3. Identify and name all substituents with their position numbers.
4. List substituents alphabetically (ignore multiplying prefixes like di-, tri-).
5. Combine: substituents + root + suffix (e.g. 2-methylbutan-1-ol).

Exam Tip: Common naming errors: (1) Not finding the longest chain that includes the functional group. (2) Numbering from the wrong end. (3) Forgetting to use commas between numbers and hyphens between numbers and letters (e.g. 2,3-dimethylbutane). Practice naming at least 10 structures before the exam.

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1.2 Key Functional Groups

Functional group	General formula	Name suffix/prefix	Example
Alkane (C-C)	C_nH_{2n+2}	-ane	Propane $CH_3CH_2CH_3$
Alkene (C=C)	C_nH_{2n}	-ene	Propene $CH_3CH{=}CH_2$
Alcohol (-OH)	$C_nH_{2n+1}OH$	-ol	Ethanol CH_3CH_2OH
Halogenoalkane (-X)	$C_nH_{2n+1}X$	chloro-/bromo-/iodo-	1-bromopropane $CH_3CH_2CH_2Br$
Aldehyde (-CHO)	$C_nH_{2n}O$	-al	Propanal CH_3CH_2CHO
Ketone (-CO-)	$C_nH_{2n}O$	-one	Propanone CH_3COCH_3
Carboxylic acid (-COOH)	$C_nH_{2n}O_2$	-oic acid	Ethanoic acid CH_3COOH
Ester (-COO-)	—	-yl -oate	Ethyl ethanoate $CH_3COOCH_2CH_3$
Amine ($-NH_2$)	$C_nH_{2n+1}NH_2$	-amine / amino-	Ethylamine $CH_3CH_2NH_2$
Amide ($-CONH_2$)	—	-amide	Ethanamide CH_3CONH_2
Nitrile (-CN)	—	-nitrile	Ethanenitrile CH_3CN
Acyl chloride (-COCl)	—	-oyl chloride	Ethanoyl chloride CH_3COCl
Acid anhydride	$(RCO)_2O$	-ic anhydride	Ethanoic anhydride $(CH_3CO)_2O$

1.3 Types of Formulae

- **Molecular formula:** Shows the actual number of each type of atom (e.g. C_3H_8O).
- **Empirical formula:** Simplest whole-number ratio of atoms (e.g. CH_2O for glucose $C_6H_{12}O_6$).
- **Structural formula:** Shows the minimal detail to make the structure unambiguous (e.g. CH_3CH_2OH).
- **Displayed formula:** Shows all bonds between all atoms.

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- **Skeletal formula:** Lines represent carbon-carbon bonds; hydrogen atoms on carbon are not shown; heteroatoms (O, N, Cl) and their hydrogens are shown.

Homologous series: A family of organic compounds with the same functional group, the same general formula, and similar chemical properties. Each successive member differs by CH_2. Physical properties (boiling point, density) show a gradual trend as chain length increases.

2. Isomerism

Isomers: Molecules with the same molecular formula but different structural arrangements of atoms. They may have different physical and/or chemical properties.

2.1 Structural Isomerism

Structural isomers have the same molecular formula but different structural formulae. There are three types:

- **Chain isomerism:** Different arrangements of the carbon skeleton (e.g. butane $CH_3CH_2CH_2CH_3$ and methylpropane $(CH_3)_3CH$, both C_4H_{10}). Branched isomers have lower boiling points because they are more compact with less surface area for London forces.
- **Position isomerism:** Same carbon chain and functional group, but the functional group is at a different position (e.g. butan-1-ol and butan-2-ol, both C_4H_9OH).
- **Functional group isomerism:** Same molecular formula but different functional groups (e.g. propanal C_2H_5CHO and propanone CH_3COCH_3, both C_3H_6O; or ethanol C_2H_5OH and methoxymethane CH_3OCH_3, both C_2H_6O).

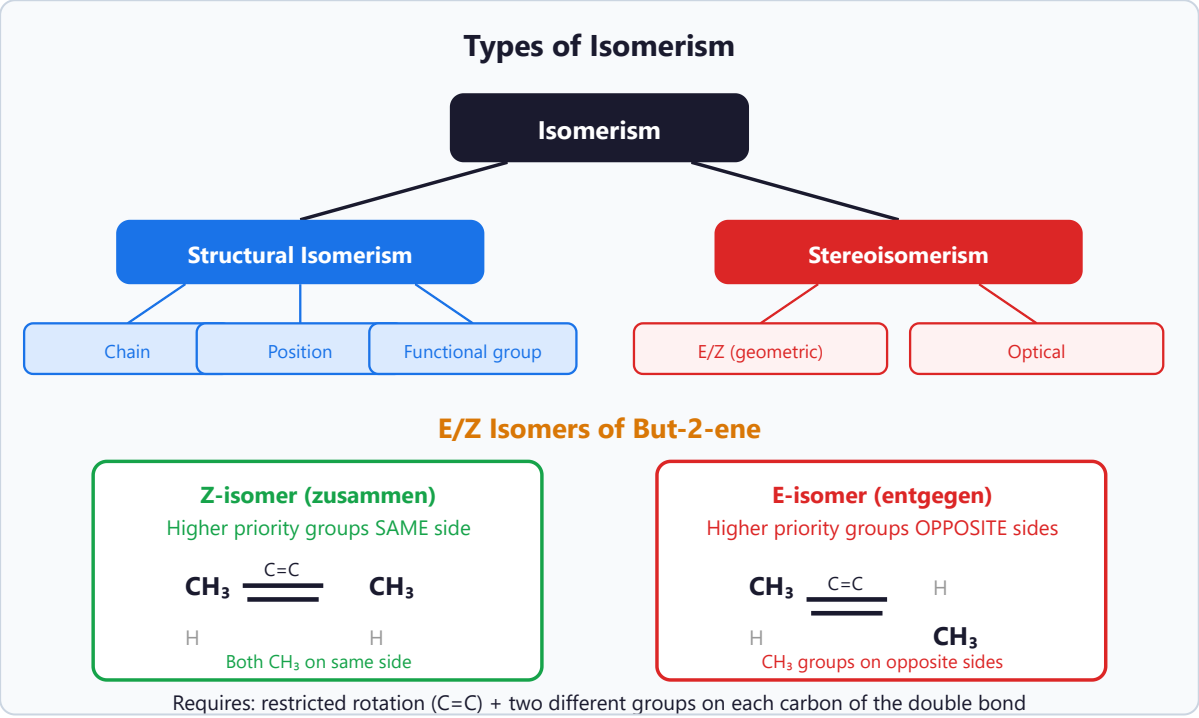
2.2 E/Z Isomerism (Geometric Isomerism)

E/Z isomerism is a form of stereoisomerism that arises when there is:

1. **Restricted rotation** about a bond (usually a C=C double bond), AND
2. **Two different groups** attached to each carbon of the double bond.

- **Z-isomer (zusammen = together):** The higher priority groups on each carbon are on the same side of the double bond.
- **E-isomer (entgegen = opposite):** The higher priority groups are on opposite sides.

Priority is assigned using the Cahn-Ingold-Prelog (CIP) rules: the atom with the higher atomic number has higher priority. If the first atoms are the same, move along the chain until a difference is found.



2.3 Optical Isomerism

Chiral centre (asymmetric carbon): A carbon atom bonded to four different groups. This produces two non-superimposable mirror image forms called **enantiomers**. A molecule with a chiral centre has no plane of symmetry.

Properties of enantiomers:

- Identical physical properties (melting point, boiling point, solubility, density).
- Identical chemical properties with achiral reagents.
- Different reactions with other chiral molecules (important in biochemistry — enzymes are chiral and usually only interact with one enantiomer).
- Rotate the plane of plane-polarised light in opposite directions: (+) enantiomer rotates clockwise (dextrorotatory), (-) enantiomer rotates anticlockwise (laevorotatory).

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Racemic mixture (racemate): A 50:50 mixture of the two enantiomers. It shows no overall rotation of plane-polarised light because the equal and opposite rotations cancel out. Racemic mixtures are produced when a new chiral centre is formed in a reaction from a non-chiral starting material (e.g. nucleophilic addition to a carbonyl).

Exam Tip: To identify a chiral centre, look for a carbon bonded to four DIFFERENT groups. Draw the two enantiomers as mirror images with a dashed line between them. In reactions that create a new chiral centre from a planar intermediate (e.g. nucleophilic addition to an aldehyde), explain that the nucleophile can attack equally from both sides, producing a racemic mixture.

3. Alkanes

3.1 Structure and Properties

Alkanes are saturated hydrocarbons containing only C-C single bonds and C-H bonds. General formula: C_nH_{2n+2}. They are relatively unreactive because:

- C-C bonds (346 kJ mol^{-1}) and C-H bonds (413 kJ mol^{-1}) are strong and require significant energy to break.
- The C-H bond is only very slightly polar (electronegativity of C = 2.5, H = 2.1), so alkanes are not attacked by nucleophiles or electrophiles.

Boiling point trends:

- Boiling points **increase with chain length** because longer chains have more electrons, leading to stronger London dispersion forces between molecules.
- Boiling points **decrease with branching** because branched molecules are more compact with less surface area, reducing intermolecular contact and weakening London forces.

3.2 Combustion

Complete combustion (excess oxygen) produces carbon dioxide and water:

$$C_nH_{2n+2} + (3n+1)/2\ O_2 \rightarrow nCO_2 + (n+1)H_2O$$

Example: $CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$ $\Delta H = -890$ kJ mol^{-1}

Alkanes are the primary component of fossil fuels (natural gas, petrol, diesel). CO_2 is a greenhouse gas that contributes to climate change by absorbing and re-emitting infrared radiation.

Incomplete combustion (limited oxygen supply) produces:

- **Carbon monoxide (CO)** — a toxic, colourless, odourless gas. It binds irreversibly to haemoglobin (forming carboxyhaemoglobin), preventing oxygen transport. CO has a higher affinity for haemoglobin than O_2.
- **Carbon/soot (C)** — particulates that cause respiratory problems, reduce air quality, and contribute to global dimming.
- **Water (H_2O)** — always produced.

Other pollutants from combustion engines:

- **NO_x (nitrogen oxides)** — formed when N_2 and O_2 react at high temperatures inside engines. Contributes to acid rain and photochemical smog.
- **SO_2** — from sulfur impurities in fuel. Causes acid rain.
- **Unburned hydrocarbons (VOCs)** — contribute to smog formation.

Catalytic converters reduce these pollutants: $2CO + 2NO \rightarrow 2CO_2 + N_2$.

3.3 Free Radical Substitution

Alkanes react with halogens (Cl_2 or Br_2) in the presence of UV light via a free radical chain mechanism. This is the only useful reaction of alkanes (besides combustion).

> **Free radical:** A reactive species with an unpaired electron, written with a dot notation (e.g. Cl•). Radicals are formed by **homolytic fission** — the equal splitting of a covalent bond, where each atom takes one electron.

Mechanism (three stages) — chlorination of methane:

1. Initiation: UV light causes homolytic fission of the Cl-Cl bond, generating two chlorine radicals:

$$Cl_2 \rightarrow 2Cl\bullet \quad \text{(UV light provides the energy)}$$

2. Propagation: Chain reactions that consume and regenerate radicals. Each step produces one radical from one radical, maintaining the chain:

$$CH_4 + Cl\bullet \rightarrow \bullet CH_3 + HCl \quad \text{(Step 1: Cl• abstracts an H atom)}$$

$$\bullet CH_3 + Cl_2 \rightarrow CH_3Cl + Cl\bullet \quad \text{(Step 2: } \bullet CH_3 \text{ reacts with } Cl_2\text{; Cl• regenerated)}$$

3. Termination: Any two radicals combine, ending the chain:

$$Cl\bullet + Cl\bullet \rightarrow Cl_2 \qquad \bullet CH_3 + Cl\bullet \rightarrow CH_3Cl \qquad \bullet CH_3 + \bullet CH_3 \rightarrow C_2H_6$$

Free Radical Substitution of Methane

Initiation — UV light breaks Cl-Cl bond by homolytic fission

Cl-Cl → Cl• + Cl• (two radicals)

Propagation (chain reaction — radicals regenerated)

Step 1: **CH_4 + Cl• → •CH_3 + HCl**

Use single-headed (fishhook) curly arrows

Step 2: **•CH_3 + Cl_2 → CH_3Cl + Cl•**

Termination (two radicals combine — chain ends)

Cl• + Cl• → Cl_2 •CH_3 + Cl• → CH_3Cl •CH_3 + •CH_3 → C_2H_6

Limitation: further substitution gives CH_2Cl_2, $CHCl_3$, CCl_4 (mixture of products)

• = unpaired electron (free radical) | Use single-headed (fishhook) curly arrows

Limitations of free radical substitution:

- **Further substitution:** The product CH_3Cl can react further to give CH_2Cl_2, $CHCl_3$, and CCl_4, producing a mixture of mono-, di-, tri-, and tetra-substituted products.

- **Multiple products from longer alkanes:** For propane, the H on C-1 or C-2 can be replaced, giving a mixture of 1-chloropropane and 2-chloropropane.
- The C_2H_6 formed in termination shows that CH_3• + CH_3• combination occurred.
- This makes free radical substitution a poor preparative method for specific halogenoalkanes.

Exam Tip: Use single-headed curly arrows (fishhooks) for free radical mechanisms to show the movement of single electrons. Use full-headed curly arrows for ionic mechanisms. You must show all three stages clearly labelled. The C_2H_6 by-product in termination is evidence for the radical mechanism (it cannot form from CH_4 and Cl_2 alone).

4. Alkenes

4.1 Structure and Bonding

Alkenes are unsaturated hydrocarbons containing at least one C=C double bond. General formula: C_nH_{2n} (for monoenes). The C=C double bond consists of:

- A **sigma (σ) bond** — formed by head-on overlap of sp^2 hybrid orbitals. This is a strong bond with electron density concentrated between the nuclei.
- A **pi (π) bond** — formed by sideways overlap of unhybridised p orbitals above and below the plane of the molecule. The π bond is weaker than the σ bond and is the region of high electron density that makes alkenes reactive.

The π electron cloud makes alkenes susceptible to attack by electrophiles (electron-pair acceptors). The C=C bond cannot rotate freely (the π bond would have to break), which is why E/Z isomerism exists in alkenes.

4.2 Electrophilic Addition

Electrophile: An electron-pair acceptor; a species that is attracted to areas of high electron density. It accepts a pair of electrons from the electron-rich π bond to form a new covalent bond. Examples: H^+, NO_2^+, $Br^{\delta+}$, carbocations (R^+).

The general mechanism for electrophilic addition to alkenes has two steps:

1. The π bond electrons are attracted to the electrophilic part of the reagent. A curly arrow shows the π bond electrons forming a new bond to the electrophile, while the bond within the electrophilic species breaks heterolytically.
2. The carbocation intermediate is rapidly attacked by the nucleophilic negative species to form the second new bond.

Key electrophilic addition reactions:

Reagent	Conditions	Product	Type of product
HBr (or HCl, HI)	Room temperature, gas bubbled through	Halogenoalkane	e.g. $CH_3CHBrCH_3$ from propene
Br_2 (in organic solvent or aqueous)	Room temperature	Dihalogenoalkane	e.g. $CH_2BrCHBrCH_3$
H_2O (steam)	H_3PO_4 catalyst, 300°C, 60-70 atm	Alcohol	e.g. ethanol from ethene (industrial method)
H_2	Ni catalyst, 150°C	Alkane	e.g. ethane from ethene (hydrogenation)
H_2SO_4 (conc.)	Room temperature, then add water and warm	Alcohol (indirect hydration)	Via alkyl hydrogen sulfate intermediate

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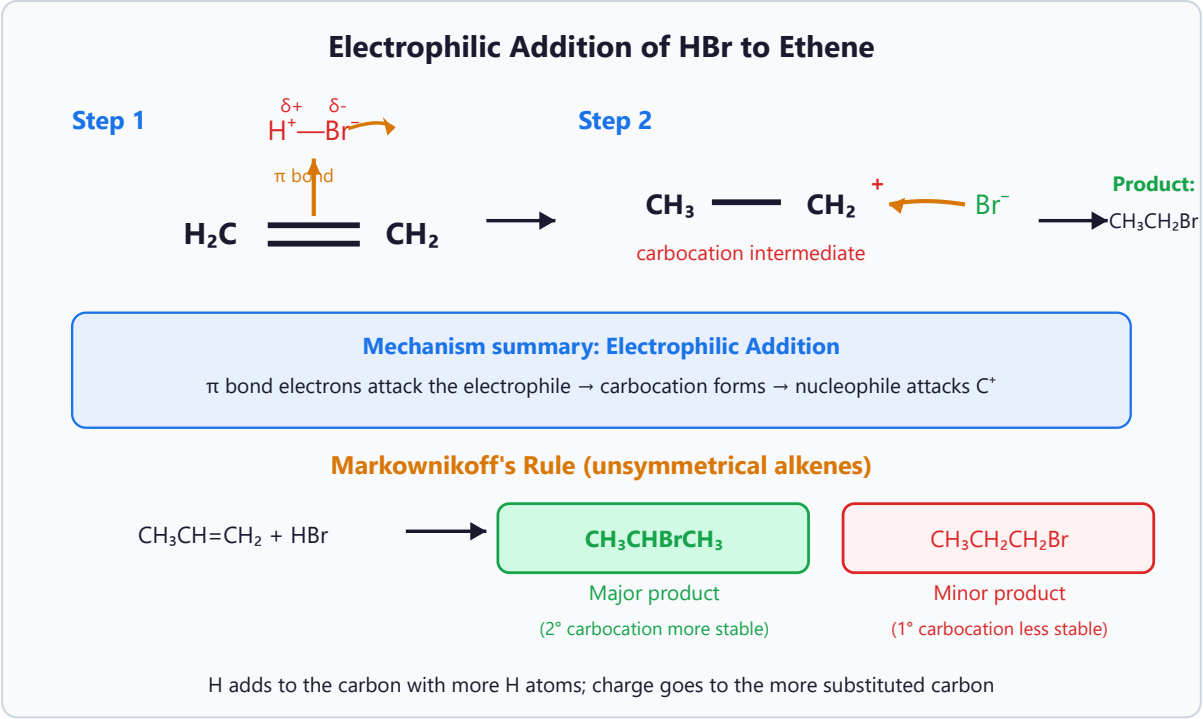


Markownikoff's rule: When HX adds across an unsymmetrical alkene (e.g. propene), the hydrogen atom adds to the carbon already bonded to more hydrogen atoms. The major product has the halogen on the more substituted carbon. This is because the intermediate carbocation is more stable when the positive charge is on the more substituted carbon (stability order: tertiary > secondary > primary). The inductive (electron-pushing) effect of alkyl groups stabilises the positive charge through hyperconjugation.

Test for unsaturation (C=C): Add bromine water (orange/brown). Alkenes decolourise bromine water immediately at room temperature without UV light. Alkanes do not decolourise bromine water (unless UV light is present, which would cause free radical substitution).

4.3 Addition Polymerisation

Many alkene monomers can join together to form a long-chain polymer by addition polymerisation. The C=C double bond in each monomer opens up, and each carbon forms a new bond to the adjacent monomer:

$$nCH_2{=}CH_2 \rightarrow -(CH_2\text{-}CH_2)_n- \quad \text{(poly(ethene))}$$

$$nCH_2{=}CHCl \rightarrow -(CH_2\text{-}CHCl)_n- \quad \text{(poly(chloroethene), PVC)}$$

Environmental issues: Addition polymers are chemically inert and non-biodegradable because of the strong C-C backbone. Disposal methods include:

- **Landfill:** Cheap but wasteful; takes up space; polymers persist for centuries.
- **Incineration:** Releases energy (can generate electricity) but produces CO_2 and possibly toxic gases (HCl from PVC).
- **Recycling:** Mechanical (melt and remould) or feedstock (crack back to monomers). Requires sorting by polymer type.

5. Halogenoalkanes

5.1 Classification and Bonding

Halogenoalkanes contain a halogen atom (F, Cl, Br, or I) bonded to a saturated carbon atom. They are classified by the number of carbon atoms bonded to the carbon carrying the halogen:

- **Primary (1°):** The halogen-bearing carbon is bonded to one other carbon (e.g. CH_3CH_2Br, bromoethane).
- **Secondary (2°):** Bonded to two other carbons (e.g. $(CH_3)_2CHBr$, 2-bromopropane).
- **Tertiary (3°):** Bonded to three other carbons (e.g. $(CH_3)_3CBr$, 2-bromo-2-methylpropane).

The C-X bond is polar (δ+C-Xδ-) because halogens are more electronegative than carbon. The δ+ carbon is susceptible to attack by nucleophiles. The polarity and bond strength of C-X bonds determine the reactivity.

5.2 Nucleophilic Substitution

> **Nucleophile:** An electron-pair donor; a species with a lone pair of electrons that is attracted to a δ+ (electron-deficient) carbon atom. It donates its lone pair to form a new covalent bond. Examples: OH^-, CN^-, NH_3, H_2O.

The nucleophile attacks the δ+ carbon, displacing the halogen as a halide ion (X^-). The mechanism depends on the classification of the halogenoalkane:

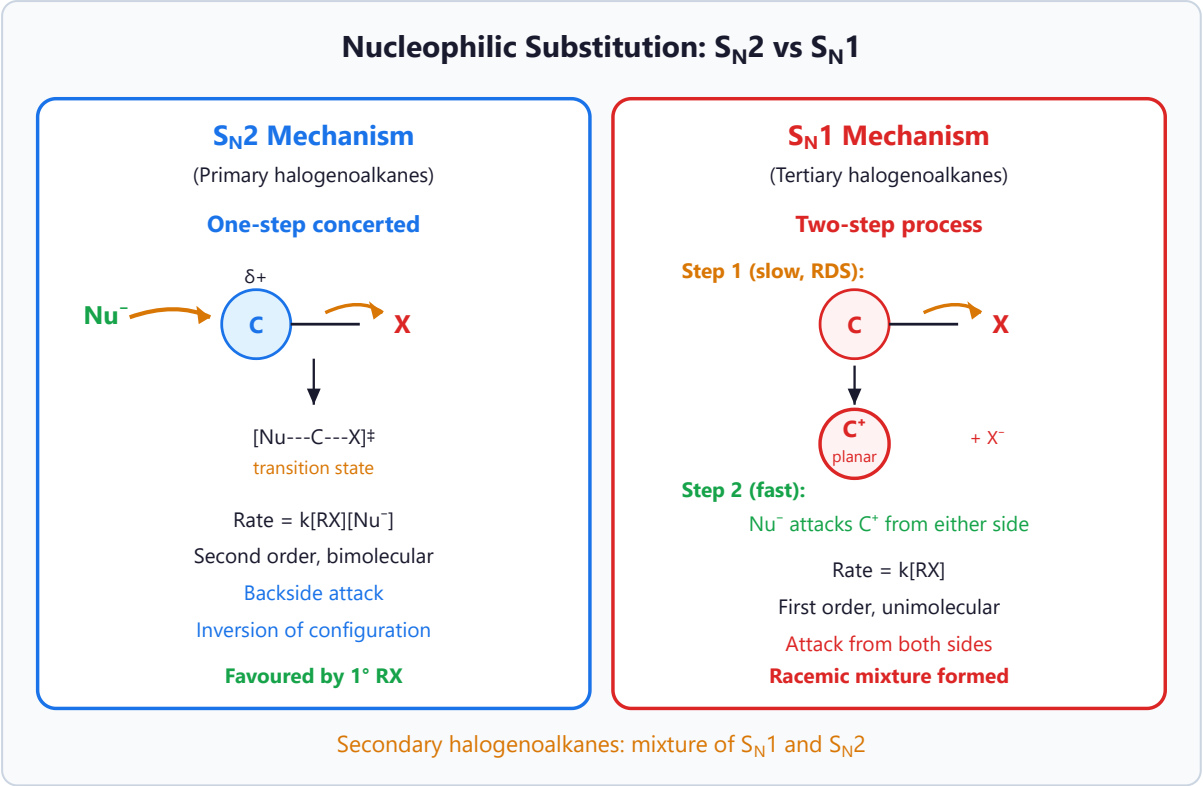
S_N2 Mechanism (primary halogenoalkanes)

- **One-step concerted mechanism.** The nucleophile attacks the δ+ carbon from the opposite side to the halogen (backside attack), while the C-X bond breaks simultaneously.
- A transition state forms where the nucleophile is partially bonded and the halogen is partially leaving: $[Nu\cdots C\cdots X]^{\ddagger}$
- **Rate equation:** Rate = k[halogenoalkane][nucleophile] — second order, bimolecular (both species in the rate-determining step).
- Produces **inversion of configuration** at the carbon centre (Walden inversion).
- Favoured by primary halogenoalkanes (less steric hindrance around the carbon).

S_N1 Mechanism (tertiary halogenoalkanes)

- **Two-step mechanism.** Step 1 (slow, rate-determining): The C-X bond breaks heterolytically to form a planar carbocation and X^-. Step 2 (fast): The nucleophile rapidly attacks the carbocation from either side.
- **Rate equation:** Rate = k[halogenoalkane] — first order, unimolecular (only the halogenoalkane in the rate-determining step).
- Tertiary carbocations are most stable (three alkyl groups donate electron density to stabilise the positive charge via the inductive effect).
- Produces a **racemic mixture** if a chiral centre is formed, because the nucleophile can attack from either side of the planar carbocation.

Secondary halogenoalkanes undergo a mixture of S_N1 and S_N2.



Key nucleophilic substitution reactions of halogenoalkanes:

Nucleophile	Conditions	Product	Carbon chain change
OH^- (from dilute NaOH/KOH)	Warm aqueous NaOH, reflux	Alcohol (R-OH)	Same chain length
CN^- (from KCN in ethanol)	Warm ethanolic KCN, reflux	Nitrile (R-CN)	Chain extended by 1 carbon
NH_3	Excess concentrated NH_3 in ethanol, sealed tube, heat	Primary amine ($R\text{-}NH_2$)	Same chain length

Why CN^- is important: Reaction with cyanide extends the carbon chain by one atom. The nitrile can then be hydrolysed to a carboxylic acid ($R\text{-}CN + 2H_2O + HCl \rightarrow R\text{-}COOH + NH_4Cl$) or reduced to an amine ($R\text{-}CN + 4[H] \rightarrow R\text{-}CH_2NH_2$ using $LiAlH_4$ or H_2/Ni).

Why excess NH_3 is used: If the amine product is formed with excess halogenoalkane still present, further substitution occurs: the amine acts as a nucleophile, leading to secondary amines, tertiary amines, and

eventually a quaternary ammonium salt. Using excess NH_3 minimises this by ensuring most halogenoalkane reacts with NH_3 rather than the amine product.

Exam Tip: In mechanism questions, show the lone pair on the nucleophile and use a curly arrow from the lone pair to the δ+ carbon. Show the bond breaking from the C-X bond to the X atom. For S_N1, clearly show two separate steps. For S_N2, show it in one step with the transition state.

5.3 Elimination

Halogenoalkanes can also undergo elimination, losing HX to form an alkene:

$$CH_3CHBrCH_3 + KOH \rightarrow CH_3CH{=}CH_2 + KBr + H_2O$$

Conditions: Hot ethanolic KOH (KOH dissolved in ethanol, heated under reflux). The OH^- acts as a **base** (not a nucleophile), removing an H^+ from a carbon adjacent to the halogen. As the C-H bond breaks, the electrons form the C=C double bond, and the C-X bond breaks to release X^-.

Substitution vs Elimination — how to control which occurs:

Factor	Favours substitution	Favours elimination
Solvent	Aqueous (water)	Ethanolic (ethanol)
Temperature	Warm	Hot (higher temperature)
Type of halogenoalkane	Primary	Tertiary
OH^- acts as...	Nucleophile	Base

5.4 Reactivity of Halogenoalkanes

The rate of nucleophilic substitution depends on the C-X bond strength (bond enthalpy), not the polarity:

C-X bond	Bond enthalpy / kJ mol^{-1}	Rate of hydrolysis
C-F	484	Very slow (strongest, hardest to break)

C-Cl	338	Slow
C-Br	276	Moderate
C-I	238	Fast (weakest, easiest to break)

Iodoalkanes react fastest because the C-I bond is the weakest and has the lowest activation energy for breaking, despite C-I being the least polar. This shows that bond strength, not polarity, determines reactivity for halogenoalkanes.

6. Alcohols

6.1 Classification and Properties

- **Primary (1°):** The -OH group is on a carbon bonded to one other carbon (e.g. butan-1-ol $CH_3CH_2CH_2CH_2OH$).
- **Secondary (2°):** The -OH is on a carbon bonded to two other carbons (e.g. butan-2-ol $CH_3CH(OH)CH_2CH_3$).
- **Tertiary (3°):** The -OH is on a carbon bonded to three other carbons (e.g. 2-methylpropan-2-ol $(CH_3)_3COH$).

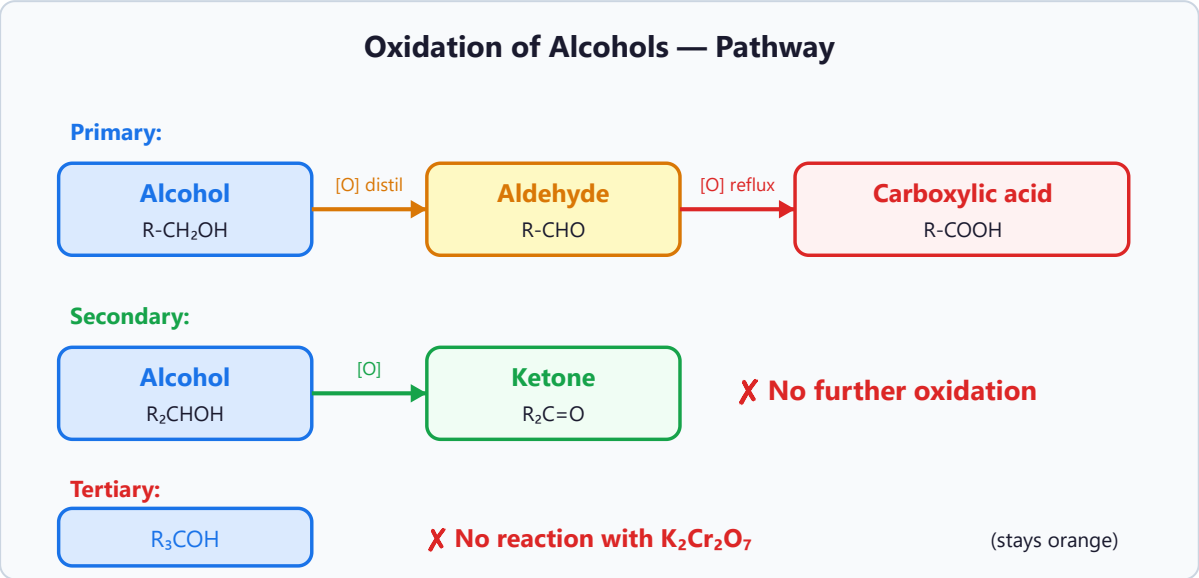
Alcohols have higher boiling points than alkanes of similar molecular mass because the -OH group forms hydrogen bonds with other alcohol molecules and with water. Short-chain alcohols are miscible with water (hydrogen bonding). As the carbon chain lengthens, the non-polar hydrocarbon part dominates and solubility decreases.

6.2 Oxidation of Alcohols

The oxidising agent is **acidified potassium dichromate(VI)** ($K_2Cr_2O_7$ / dilute H_2SO_4). When oxidation occurs, the orange $Cr_2O_7^{2-}$ is reduced to green Cr^{3+}.

Alcohol type	With distillation	Under reflux
Primary (1°)	Aldehyde (removed as it forms, preventing further oxidation)	Carboxylic acid (aldehyde oxidised further)

Secondary (2°)	Ketone (cannot be oxidised further — no H on the carbonyl C)
Tertiary (3°)	No reaction — dichromate stays orange (no H on the C-OH carbon to remove)



Equations using [O] notation:

$$\text{Primary: } CH_3CH_2OH + [O] \rightarrow CH_3CHO + H_2O \ \ \text{(distillation)}$$

$$\text{Then: } CH_3CHO + [O] \rightarrow CH_3COOH \ \ \text{(reflux)}$$

$$\text{Secondary: } CH_3CH(OH)CH_3 + [O] \rightarrow CH_3COCH_3 + H_2O$$

Exam Tip: The key distinction between distillation and reflux: Distillation removes the aldehyde product as soon as it forms (it has a low boiling point and evaporates). Reflux keeps the mixture at the boiling point without losing volatile components, so the aldehyde is further oxidised to the carboxylic acid. You must specify which apparatus is used when answering questions about oxidation of primary alcohols.

6.3 Dehydration (Elimination)

Alcohols can be dehydrated to form alkenes by removing H_2O:

$$CH_3CH_2OH \rightarrow CH_2{=}CH_2 + H_2O$$

Conditions: Concentrated H_2SO_4 or concentrated H_3PO_4 at elevated temperature (170°C for ethanol). Alternatively, pass alcohol vapour over heated aluminium oxide (Al_2O_3) at 300-400°C. This is an elimination reaction — the acid acts as a catalyst.

6.4 Esterification

An alcohol reacts with a carboxylic acid in the presence of a concentrated H_2SO_4 catalyst to form an ester and water (condensation reaction):

$$CH_3COOH + CH_3CH_2OH \rightleftharpoons CH_3COOCH_2CH_3 + H_2O$$

This is a reversible equilibrium reaction. Conditions: warm with concentrated H_2SO_4 catalyst under reflux. The yield can be improved by using excess of one reactant or removing the water.

6.5 Substitution to Halogenoalkanes

Alcohols can be converted to halogenoalkanes by replacing the -OH group:

- **With HCl, HBr, or HI:** e.g. $CH_3CH_2OH + HBr \rightarrow CH_3CH_2Br + H_2O$ (NaBr + conc. H_2SO_4 generates HBr in situ)
- **With PCl_3 or PCl_5:** $3ROH + PCl_3 \rightarrow 3RCl + H_3PO_3$
- **With $SOCl_2$ (thionyl chloride):** $ROH + SOCl_2 \rightarrow RCl + SO_2 + HCl$ (preferred method as by-products are gases that leave the mixture)

6.6 Combustion

Alcohols burn in excess oxygen: $C_2H_5OH + 3O_2 \rightarrow 2CO_2 + 3H_2O$. Ethanol is used as a biofuel, either neat or blended with petrol (E10 = 10% ethanol). It is considered closer to carbon neutral than fossil fuels because the CO_2 released was originally absorbed by the crops used to make the ethanol (via fermentation of sugars: $C_6H_{12}O_6 \rightarrow 2C_2H_5OH + 2CO_2$, using yeast at 25-37°C).

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7. Organic Analysis

7.1 Mass Spectrometry

The **molecular ion peak** (M^+) in a mass spectrum gives the relative molecular mass of the compound. The molecular ion forms when the molecule loses one electron in the ionisation chamber: $M \rightarrow M^{+\bullet} + e^-$.

Fragmentation: The molecular ion can break apart into smaller fragments. The m/z values of fragment ions help deduce the structure. Common fragment losses:

m/z loss	Fragment lost
15	$CH_3\bullet$ (methyl group)
17	$OH\bullet$ (hydroxyl group)
18	H_2O (from alcohols/carboxylic acids)
28	CO (from aldehydes/ketones) or C_2H_4
29	$CHO\bullet$ or $C_2H_5\bullet$
31	$CH_2OH\bullet$ (from primary alcohols)
45	$OC_2H_5\bullet$ (from ethyl esters)

The **M+1 peak** (due to ^{13}C) can help determine the number of carbon atoms in the molecule. The height of M+1 as a percentage of M is approximately 1.1 × (number of C atoms).

7.2 Infrared Spectroscopy

Different covalent bonds absorb infrared radiation at characteristic frequencies (wavenumbers, cm^{-1}). IR spectroscopy identifies functional groups present in a molecule.

Bond	Wavenumber / cm^{-1}	Appearance	Found in

O-H (alcohol, hydrogen-bonded)	3230-3550	Broad absorption	Alcohols, phenols
O-H (carboxylic acid)	2500-3300	Very broad absorption	Carboxylic acids
N-H	3300-3500	Medium, may show 2 peaks (NH_2)	Amines, amides
C-H	2850-3100	Strong	Most organic compounds
C=O	1680-1750	Strong, sharp	Aldehydes, ketones, carboxylic acids, esters, amides
C=C	1620-1680	Medium	Alkenes
C-O	1000-1300	Strong	Alcohols, esters, ethers

The **fingerprint region** (below ~1500 cm^{-1}) is unique to each molecule, like a molecular fingerprint. Computer databases can match an unknown spectrum to a known compound in this region.

Greenhouse gases and IR: CO_2, H_2O, and CH_4 absorb IR radiation because their bonds can change dipole moment when they vibrate (asymmetric stretching and bending). N_2 and O_2 do not absorb IR because they are non-polar and have no change in dipole during vibration.

Exam Tip: You do NOT need to memorise exact wavenumbers — they are given in the data booklet. Focus on recognising: (1) the broad O-H absorption in alcohols vs the very broad O-H in carboxylic acids, (2) the strong sharp C=O peak, (3) the absence of O-H after esterification. When interpreting spectra, look at the absorption shape and position, not just whether a peak exists.

7.3 Test-Tube Reactions for Functional Group Identification

Test	Reagent and conditions	Positive result	Identifies
Bromine water test	Add Br_2(aq), shake at room temperature	Orange → colourless	Alkene (C=C double bond)

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Acidified $K_2Cr_2O_7$	Warm with H_2SO_4	Orange → green	Primary or secondary alcohol (or aldehyde)
Tollens' reagent	$[Ag(NH_3)_2]^+$ in NaOH, warm gently in water bath	Silver mirror on glass	Aldehyde (not ketone)
Fehling's solution	Cu^{2+} in NaOH/tartrate, heat	Blue → brick-red precipitate (Cu_2O)	Aldehyde (not ketone)
Na_2CO_3 / $NaHCO_3$	Add to unknown compound	Effervescence (CO_2 gas)	Carboxylic acid
2,4-DNPH (Brady's reagent)	Add to unknown, shake	Orange/yellow precipitate (hydrazone)	Aldehyde or ketone (C=O present)
PCl_5	Add to unknown at room temperature	White misty fumes of HCl	-OH group (alcohol or carboxylic acid)
Iodoform test (tri-iodomethane)	NaOH(aq) + I_2(aq), warm	Yellow precipitate of CHI_3 (antiseptic smell)	Methyl ketone (CH_3CO-) or secondary alcohol with $CH_3CH(OH)$-

Exam Tip: A common multi-step identification question: (1) Use 2,4-DNPH to confirm C=O present. (2) Then use Tollens' reagent to distinguish aldehyde (silver mirror) from ketone (no reaction). (3) If the compound is an alcohol, use $K_2Cr_2O_7$ to test for primary/secondary (colour change) vs tertiary (no change). (4) The iodoform test is positive for methyl ketones and methyl secondary alcohols specifically — it is very commonly examined.

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